The Missing Middle Managers: Labor Costs, Firm Structure, and Development[*][†]

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Abstract

This paper establishes that the high cost of middle management is an important barrier to the expansion of the modern sector in developing countries. We use data from compensation and recruitment consulting firms to show that large, leading firms pay $42,000 per year for managers and business professionals even in the poorest countries in the world. Managers and business professionals also account for a large share of total costs in modern firms. We use an appropriate technology model to show that cost of management has a large impact in absolute terms and relative to other factors studied in the literature.

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1 Introduction

Developing and emerging economies are characterized by a dual economy where productive modern firms using new technologies coexist with small, unproductive traditional firms or own-account workers who produce using out-of-date technologies.[1] A central question for economics is why the modern sector does not expand and displace the traditional sector, either through domestic adoption of new technologies (Murphy *et al.*, 1989; Ciccone, 2002; Cole *et al.*, 2016) or foreign direct investment by multinational firms (Antràs & Yeaple, 2014; Ramondo *et al.*, 2015).

This paper points to a factor that has received little attention in the literature: the high cost of middle management in poor economies. That running a modern economy requires a large number of skilled workers such as managers and business professionals has been known since the emergence of development economics. This insight was a key motivation for the emphasis placed on human capital investment in early growth plans across the developing world. However, the cost of skilled labor has not looked like a promising predictor of modernization since educational wage premia are relatively similar across countries at different income levels (Banerjee & Duflo, 2005; Rossi, 2022).

The main contribution of this paper is to show that we get a different picture when we focus on the wages actually paid by modern firms rather than looking at general educational or white-collar wage premia. Modern firms face surprisingly high costs for middle managers and business professionals in developing countries, and we show that this is an important barrier to the expansion of new technologies and re-organization of production into large firms.

We start with the data. A key challenge with studying dual economies empirically is that standard, representative data sets contain information from workers or firms in both the traditional and modern sectors, which makes it hard to isolate the costs facing the modern sector.[2] This matters if the modern sector uses different types of workers and faces different costs than the average firm in the economy. For example, Bassi *et al.* (2023) show that managers and production workers in traditional firms perform similar tasks; this is unlikely to be the case for, say, multinational firms. We overcome this challenge by drawing on data from two consulting companies that help modern firms navigate

[1] The notion of the dual economy dates to Lewis (1954) and Jorgenson (1961). See Comin & Hobijn (2010) for empirical evidence on lags in technology adoption, Gollin (2008) on the prevalence of self-employment, and Poschke (2018), and Bento & Restuccia (2021) for facts about firm and establishment size.

[2] This is a common challenge in taking dual economy models to the data (e.g., Buera & Trachter, 2024).

labor markets for skilled workers in developing and emerging economies. The first is a compensation consulting firm, which advises clients on labor market conditions and pay. We have access to this firm's database containing records of actual compensation paid by over 1,000 of the firm's clients to over 300,000 workers in 146 countries around the world.[3] The second is a recruitment consulting firm. We have access to this firm's published salary survey, which provides less exhaustive information on the prevailing salaries paid in skilled labor markets based on their market expertise.

The key observation is that firms face high costs for managers and business professionals, even in the poorest countries. The database of the compensation consultant shows that the average annual compensation for managers and business professionals is $42,000 in the poorest decile of countries in the sample, as compared to $80,000 in the richest decile. The salary guide of the recruitment consultant lists similarly high salary ranges: the midpoint of the annual salary range for a General Manager in Central Africa as $101,000; for a General Accountant in East Africa, $35,000; for a Sales & Marketing Manager in West Africa, $79,000.

Our second contribution is to quantify the importance of the cost of middle management for the joint decision to adopt new technologies and re-organize production into large firms. We develop an appropriate technology model as in Basu & Weil (1998) or Acemoglu & Zilibotti (2001) where potential producers choose between traditional or modern production. Modern production conveys the advantage of producing at scale with new technologies. Producers weigh this benefit against the difference in factor costs associated with switching to modern production. We show that factors can explain low adoption if they are relatively expensive in developing countries and used more intensively by modern than by traditional production.

We estimate the importance of management in modern and traditional firms using data from the literature on firm hierarchies (Caliendo *et al.*, 2015). We combine this with the data on the cost of management to show that the high price of management has a deterring effect equivalent to a 21 percent tax levied on the gross output of modern firms in developing countries. This finding is unusual in the context of the appropriate technology literature; traditional work focused on the potential role of scarcity of educated

[3] Hjort *et al.* (2020) use the same database to show evidence that the low variation in compensation across countries within multinational employers in part reflects headquarter wages themselves affecting foreign establishment wages. Quantitatively such a direct effect can only explain a fraction of the bigger, multi-source, middle-manager-market phenomenon we study in this paper. For example, we show that leading domestic firms also pay a similar level of compensation.

workers, but empirically the relative cost of educated labor does not vary much with development. We also use our framework to quantify the effect of electricity and financing costs and find that management costs are two to four times as large a deterrent.

We conclude that labor costs deserve greater attention as a barrier to modernization. Many barriers studied in the literature, such as financial constraints, poor contracting institutions, weak property rights, unreliable electricity, or trade barriers, reduce the demand for skilled managers, and should ceteris paribus lower management prices Our finding of high prices suggests that such explanations need to be complemented with theories focusing on skill supply and the functioning of skilled labor markets in poor countries.

In addition to the work cited so far, our paper is also related to a small set of papers that document little variation in pay across countries for highly specialized, skilled workers (Hjort *et al.*, 2020; Minni, 2024).[4] Our contribution is to show that this pattern holds both within and between firms and to describe the effect of these costs on firms' behavior. We also touch on the growing literature demonstrating the importance of management (Bloom *et al.*, 2014). Our findings on relative costs help rationalize why firms choose low-quality management, including the widespread use of family members as managers, instead of hiring professional management (Bloom *et al.*, 2013).

2 Data

Our data come from two consulting companies that help large, leading firms navigate labor markets for skilled workers in developing and emerging economies. The first is a compensation consulting firm that advises clients on how their pay at an establishment compares to the local market. During this process, the compensation consultant collects data on pay of all currently employed workers in the client's establishment. The second is a recruitment consulting firm that helps clients find and hire suitable workers. During this process, the recruitment consultant develops data on the going market rate for newly hired workers in key positions. We describe both sources in turn.

Our agreement with the global compensation consulting company prevents us from revealing their name, so we refer to them as the "Company". Their central business proposition is to provide clients with information on how the compensation of their employees compares with the prevailing rate for similar workers in the local labor market. The

[4] By contrast, Brinatti *et al.* (2022) find more variation in pay for workers who use an internet job platform.

Company's niche among compensation consulting firms is information on developing and emerging markets.

The Company employs professional jobs analysts who conduct interviews to learn about the tasks, responsibilities, and skills associated with each position. The analysts use this information to translate each position into the Company's internal, globally standardized job classification scheme. This scheme is detailed, consisting of more than 200 job titles that allow for both horizontal and vertical differentiation of jobs (accounting versus human resources; junior accountant versus senior accountant). This work is invaluable for our purposes because it means that the data on compensation for the same job across countries is much more comparable than that produced by the standard method, which involves economists applying crosswalks to workers' self-reported occupations.

The Company bases market compensation on the data provided by past clients in the same labor market. After providing the client comparisons of its pay to this benchmark, the Company adds the client's data to the Company's database for future comparisons. We have access to the database as of late 2015, which in turn reflects compensation reported by clients spanning the years 2000–2015. Each observation reports the firm name, city and country, year, standardized job classification, average compensation of workers in the position in the establishment, and in many cases also the total number of such workers. All observations pertain to local workers; expatriates are reserved to a separate database, which unfortunately we cannot access.

We use the firm name to merge on the firm's industry, profit/non-profit status, and headquarters location. Throughout, we restrict attention to for-profit firms and exclude charities and governmental organizations. A central feature of the database for our research question is that it covers almost exclusively modern business enterprises. Three-fourths of the compensation observations come from multinational firms. These firms are based primarily in North America (predominantly the United States), followed by Africa and Europe. Many firms in the database are large, well-known, publicly listed companies. To this point, the publicly listed U.S. firms in the database account for 32 percent of all revenue and 44 percent of all R&D investment in Compustat North America. The remaining one-fourth of observations come from large domestic firms. Both types of firms come from a wide variety of sectors, including banking, consulting, health care, mining and other natural resources, technology, telecommunications, and transport.

The establishments that appear in the Company database provide local business and headquarter services. We have verified that many firms also have separate production or sales establishments in the same country, but these establishments are not in the database, likely reflecting that the labor markets for sales and production workers are thicker, information on prevailing compensation is easier to access, and compensation for such workers is much lower. The distribution of occupations is heavily weighted towards managers and business professionals, with a small share of support workers who are captured incidentally because they work in the local headquarters establishment; see Appendix A for details.

We measure compensation as total gross pay, which is the sum of gross wage, gross bonus, and other gross income. All amounts are reported in contemporaneous U.S. dollars. We convert all earnings back into local currency units using contemporaneous market exchange rates. We then adjust all amounts to year 2017 local currency units by adjusting for the average rate of nominal wage growth between year t and year 2017, inferred from the growth rate of nominal GDP per worker. This adjustment makes salaries comparable over time by assuming that each occupation would have experienced the aggregate average wage growth; it misses any occupation-specific wage growth. Finally, we convert year 2017 wages in local currency units to year 2017 international dollars using the PPP exchange rate.[5] We trim the bottom and top 0.5 percent of the real earnings distribution, which eliminates some outliers that look to be the result of miscoding.

Our second data source is information provided by the recruitment consultant Robert Walters, a self-described "global specialist professional recruitment consultancy."[6] Robert Walters helps firms recruit for positions in key business areas that overlap substantially with the labor markets covered by the Company. As a part of their business, they employ recruiters who identify and maintain contacts with workers who are interested in moving to new positions. When contacted by clients, they use this information to help fill vacancies.

Like most recruitment consultants, Robert Walters charges clients a fee that is based on the compensation the new hire receives. In most cases, the fee structure is a fixed percentage of the first year's salary, exclusive of benefits. Thus, as part of its business Robert Walters amasses a wealth of information on the actual first-year salaries paid to newly-hired workers in specialized labor markets. It uses this information to produce an annual Salary Survey, which is what we access for data. The Salary Survey aggregates the information in Robert Walters' database to provide a salary range for key positions by

[5] All data for the adjustments from World Bank (2022).

[6] https://www.robertwaltersgroup.com/careers/robert-walters/where-we-work.html, July 18, 2023.

broad regions. For example, it reports the typical salary range for HR Managers in West Africa over the previous few years. While this aggregation prevents us from merging on firm characteristics or conducting detailed investigation, it is useful to have data from a second, publicly available source.

We focus on their data for Africa exclusive of South Africa, which contains most of the poorest countries in the Company's sample. The geographic detail in the Salary Survey increases over time; we collect data from the 2017 survey, which was the first to decompose Africa into four geographic regions: North Africa, East Africa, West Africa, and Central-South Africa (Robert Walters, 2017). The Salary Survey includes a salary range for 65 roles spread across these four regions. We replace the salary range for each position with the midpoint and adjust to 2017 international dollars using the same algorithm that we applied to the Company's database.

3 Empirical Results

We now turn to what these two data sets reveal about the cost of management and business professionals around the world. We start by computing the average compensation (across all job titles) for each country in the Company database. The average compensation figures are high, even in developing countries. For example, the poorest decile of countries have an average GDP per worker of just over \$4,000 but report an average compensation per worker of \$48,720. Robert Walter's data include similarly high figures for the poorest countries in the world. Since its salary survey is public, we can report precise figures: the midpoint of the annual salary range for a General Manager in Central Africa as \$101,000; for a General Accountant in East Africa, \$35,000; for a Sales & Marketing Manager in West Africa, \$79,000. Further, the two data sets broadly agree on salary levels, with Robert Walters reporting salaries that are roughly 30 percent higher for the same jobs in Africa. This gap is plausibly accounted for by the fact that Robert Walters data deals exclusively with newly-hired workers who may have higher salaries.

Thus, both data sources agree that large, modern firms face a high cost of managers and business professionals in developing countries. In the rest of this section, we use the greater detail in the Company's database to help understand and decompose this result. We estimate regressions of the form

$$\log(w_{c,t,f,j}) = \gamma + \eta \log(y_c) + \beta X_{c,t,f,j} + \varepsilon_{c,t,f,j}, \tag{1}$$

where $w_{c,t,f,j}$ is the total real gross compensation for workers in country c and year t working for firm f in standardized job j, y_c is the real GDP per worker in country c, and X is a vector of controls. The main parameter of interest is η, the elasticity of compensation with respect to GDP per worker.

This compensation elasticity captures how much the cost of management for modern firms varies with development. Two simple benchmarks can help build intuition. The first is a standard neoclassical growth model with homogeneous labor. A representative firm in each country takes input costs as given and produces output using a Cobb-Douglas production function with country-specific total factor productivity. In this model, compensation per employee is the labor share times GDP per worker, which implies that the compensation elasticity is one. The second benchmark is a simple application of the law of one price with heterogeneous labor. If a given type of worker earns the same compensation in all countries, then the compensation elasticity is zero.

Table 1 shows the results from estimating equation (1). Recall that each observation in our database includes the number of workers and average compensation per country-year-firm-job; we weight the regression by the number of workers and report robust standard errors. Column (1) shows the simplest specification, which includes no controls at all. In this case, the estimated elasticity is 0.16. Appendix Figure A.2 plots this average compensation against PPP GDP per worker for all possible countries. Columns (2)–(5) show the effect of adding fixed effects and interactions for year, job, and firm, which reduce the elasticity by about half. Column (5) is particularly useful for alleviating any remaining concern about the comparability of jobs across countries, as it compares compensation for the same job in the same parent firm across affiliates in different countries.

TABLE 1: ESTIMATED COMPENSATION ELASTICITY W.R.T. GDP PER WORKER

	(1)	(2)	(3)	(4)	(5)
Log GDP p.w.	0.158***	0.114***	0.113***	0.088***	0.085***
	(0.038)	(0.007)	(0.007)	(0.005)	(0.004)
Fixed Effects	None	Year + Job	Year × Job	Year + Job + Firm	Year × Job × Firm
R-squared	0.021	0.718	0.727	0.842	0.853
N	160,681	160,656	160,455	160,653	85,062

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

We investigate the heterogeneity of this result along two dimensions. First, we consider whether it differs much between foreign affiliates of multinational firms and domestic establishments, inferred from whether an establishment is in the same country as the firm's headquarters. The results are shown in Table 2. We cannot include firm fixed effects when investigating domestic establishments, so we control for job-year interactions as in column 3 of Table 1. The first column repeats those results for comparison.

TABLE 2: ESTIMATED COMPENSATION ELASTICITY BY ESTABLISHMENT TYPE

	All	By Firm Type	
		Foreign	Domestic
Log GDP p.w.	0.113***	0.111***	0.110***
	(0.007)	(0.006)	(0.014)
Fixed Effects	Year × Job	Year × Job	Year × Job
R-squared	0.727	0.732	0.727
N	160,455	126,039	34,161

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

The remaining two columns show the results for foreign affiliates and domestic establishments. Note again that the majority of our sample is foreign affiliates. However, the estimated compensation elasticity for the two groups is almost identical. This implies that our findings also apply to large, modern domestic firms.

We also investigate how our results vary by skill level. We use the vertical dimension of the Company's internal job classification scheme to group workers into four broad skill levels. The bottom skill level includes workers who are not in manager or business professional roles. These are cleaners, guards, drivers, and so on. The remaining groups capture different skill levels of managers and business professionals. The low skill level includes workers with clerical jobs, such as secretaries. The medium skill level includes workers with business associate and business professional jobs, such as accountant. The high skill level includes those with upper management role, such as senior executive.

Table 3 shows the implied compensation elasticity for these different skill groups, each estimated with job-year interactions, which control for heterogeneity across countries in the mix of jobs within each broad group. The first column again shows that the elasticity in the aggregate is 0.11. Turning to the results by skill level, there is a very clear pattern: the elasticity is lower for workers with higher skill levels. While the elasticity is 0.21 for the non-management workers, it falls to 0.15 for the least-skilled managers, 0.07 for the

TABLE 3: ESTIMATED ELASTICITY OF COMPENSATION BY SKILL LEVEL

	All	By Skill Level			
		Non-Management	Low	Medium	High
Log GDP p.w.	0.113***	0.205***	0.145***	0.069***	0.012*
	(0.007)	(0.019)	(0.013)	(0.005)	(0.005)
Fixed Effects	Year × Job	Year × Job	Year × Job	Year × Job	Year × Job
R-squared	0.727	0.364	0.467	0.251	0.165
N	160,455	10,322	71,111	47,090	31,932
Example Job		Driver	Secretary	Accountant	Senior Executive

Standard errors in parentheses
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

medium-skilled managers, and 0.01 – essentially zero – for the high-skilled managers.[7] This finding is consistent with recent research by Minni (2024), who documents heterogeneity in pay patterns by skill level within a large multinational firm.

The low compensation elasticity for managers and business professionals – equivalently, higher relative compensation for managers and business professionals in developing countries – is the central empirical finding of our paper. Our next goal is to show that these costs are a quantitatively significant deterrent to firms considering entering developing countries.

4 Quantifying the Importance of Management Costs

In this section we develop an appropriate technology model that can be used to quantify the importance of factor costs for the adoption or expansion of large, modern firms. We show that factors are important deterrents to such expansions or spreads if they are relatively more expensive in developing countries and used relatively more intensively in modern production. We use this insight to quantify the importance of management costs and compare it to several other factors considered in the literature, such as electricity or financing costs.

[7] In Appendix A.3 we show that there is no evidence that clients vary their hiring patterns in response to these cost differentials.

4.1 Appropriate Technology Framework

We assume that the economy has a unit interval of goods $i \in [0,1]$ and a set of factors $f = 1, \ldots, F$. Each good can be produced using a modern and a traditional technology, with production functions

$$y^T(i) = A^T(i) \prod_{f=1}^{F} \left(\frac{x_f^T(i)}{\alpha_f^T} \right)^{\alpha_f^T},$$

$$y^M(i) = e^{-\tau} A^M(i) \prod_{f=1}^{F} \left(\frac{x_f^M(i)}{\alpha_f^M} \right)^{\alpha_f^M},$$

where $x_f^T(i), x_f^M(i)$ denote factor inputs and α_f^T, α_f^M denote factor intensities. The terms $A^T(i), A^M(i)$ encode the productivities of different varieties, and capture that the relative productivity of modern and traditional technologies might differ depending on the good produced. For example, there are greater economies of scale in the production of cement than in plumbing services. The term τ captures general barriers to modern production such as contracting frictions, bad intellectual property rights, or low-quality infrastructure.

We think of the traditional technology as the one that was used historically. The modern technology can be introduced either through adoption by local firms or through foreign direct investment by multinationals. In order for this adoption or investment to occur for good i, it must be the case that the modern technology offers an advantage over the traditional technology in terms of lower unit costs.

Without loss of generality, we assume that $A^M(i)/A^T(i)$ is decreasing in i, which implies that there is a cut-off technology i^*

$$\frac{A^M(i^*)}{A^T(i^*)} = e^{\tau} \prod_f w_f^{(\alpha_f^M - \alpha_f^T)}, \tag{2}$$

where w_f is the price of factor f. Intuitively, adoption is low if there are high barriers τ, or if there are high factor prices w_f of factors that are used intensively in modern production compared to in traditional production. Given the adoption equation, the following is a sufficient statistic for the adoption effect of a difference in factor prices:

$$\sum_f (\alpha_f^M - \alpha_f^T) \Delta \log w_f.$$

4.2 Quantifying the Role of Factor Prices

We use this equation to quantify the role of differences in three sets of factor prices: management, electricity, and financing. In order to perform these calculations, we need data on both cross-country differences in relative factor prices $\Delta \log w_f$ as well as differences in factor shares between modern and traditional $\alpha_f^M - \alpha_f^T$.

Management prices. We take cost of management from the Company's database, using wages of managers and business professionals residualized for job-year interactions to control for differences in workforce composition. We estimate the cost of production and supervisory labor for a wide range of countries by taking 44 percent of each country's non-agricultural GDP per worker, which we show in Appendix A.1 closely approximates the earnings of such workers in several countries where we have microdata. The ratio of the two implies that the relative cost of management is 14 times higher in the poorest countries in our sample than in the richest.

For compensation shares of management, we turn to the literature that investigates firms and their workers in terms of their place in a production hierarchy (Garicano & Rossi-Hansberg, 2006). Caliendo *et al.* (2015) use French matched employer-employee data that describes each worker's position in the firm hierarchy, from production workers to top managers. They show that firms follow a natural hierarchy: simpler firms have one or two layers (production workers and supervisors), while more complex firms have three or four layers (middle and upper managers). We define modern firms as those with three or four hierarchical layers, and the manager share as the compensation share of workers at levels three or four.[8]

We find the compensation share of managers at modern firms by multiplying their reported compensation share of payroll (0.28) with a labor share of value added of 0.66 and intermediate input share of 0.5, yielding $\alpha_{man}^M = 0.09$. The approach implies $\alpha_{man}^T = 0$ by construction, as we have defined traditional firms as those with only production workers and supervisors, but no middle or upper management.

Electricity prices. For electricity, we model a price reduction equivalent to fully replacing generator-produced electricity with grid-based electricity. We base our numbers on the study by Fried & Lagakos (2023) of electricity use by firms in Sub-Saharan Africa.

[8] This division implies that just over half of French firms are modern, but their larger size means that they account for 95 percent of value added.

They find that firms with generators produce 59% of their electricity at 5.51 times the average cost of grid-based electricity (Fried & Lagakos, 2023, Online Appendix, Table C.1). The higher costs reflect both variable costs (\$0.28/kWh vs \$0.06/kWh) and maintenance and capital costs for generators (\$0.08/kWh), all in 2014 dollars.

We define the price shock of eliminating generator dependence as:

$$\Delta \log w_{electricity} = \log \left(\frac{0.06}{0.06 \times (1 - 0.59) + 0.59 \times 0.34} \right) = -1.32.$$

For the electricity factor share, we set $\alpha^M_{electricity}$=0.035, derived by multiplying Lagakos and Fried's estimate of electricity's value-added share in the modern sector (0.07) with an intermediate input share of 0.5. Consistent with their assumption that traditional firms do not use electricity, we set $\alpha^T_{electricity} = 0$.

Financing costs. To model financial frictions, we examine the impact of reduced credit spreads on factor prices facing modern firms. Our experiment considers a fall in credit spreads from 45 to 5 percentage points. This calibration draws on Cavalcanti *et al.* (2023), who analyze a comprehensive loan-level data from a credit registry in Brazil. The initial 45 percentage point spread represents the credit-weighted average of credit spreads in their study, while the 5 percentage point endpoint is an estimate of US credit speads based on the difference between the US high-yield index and the market yield on 10-year U.S. treasury securities (2015 values).

To calculate the resulting decrease in capital costs, we follow Cavalcanti *et al.* (2023) in assuming a risk-free rate of 2.5% and a depreciation rate of 3%, implying a reduction in capital user costs $\log \frac{0.025+0.03+0.05}{0.025+0.03+0.45} = -1.57$. For the cost share of external financing in the modern sector, we set $\alpha^M_{ext} = 0.085$. This estimate combines three factors: a capital share of value added (0.33) from Cavalcanti et al., an intermediate input share of gross output (0.5), and an estimated external financing share in modern firms (0.4). Our external financing share estimate is an upper bound based on Cavalcanti et al.'s finding that 19% of the capital stock is externally financed. Conservatively assuming no external financing for traditional firms and at least 50% of the aggregate capital stock in modern firms (as a comparison, the top 10% of firms have 77% of employment), we deduce that the external financing share in modern firms cannot exceed 0.4. Finally, we set the share of external financing in traditional firms to zero. This choice maximizes the difference to the modern sector, creating a best-case scenario for financial frictions to influence the adoption of modern technologies.

4.3 Results

Table 4 reports the effect on total costs to operate a modern firm for each factor. The cost of management has a large effect, equivalent to a gross output tax or wedge of $1 - \exp(-0.24) = 22$ percent. This a considerably larger effect than that associated with electricity or financing. This result stems from the fact that management has both a large cross-country relative price difference and a large difference in factor intensity between modern and traditional production. By contrast, electricity has the smallest effect because it has a substantial price difference but a small cost share, limiting the overall potential for savings.

TABLE 4: EFFECT OF FACTOR PRICES

	Factor		
Term	Management	Electricity	Financing
$\Delta \log w_f$	-2.64	-1.32	-1.57
α_f^M	0.09	0.035	0.085
α_f^T	0	0	0
$(\Delta \log w_f) \times (\alpha_f^M - \alpha_f^T)$	-0.24	-0.046	-0.13

We consider several robustness checks on these results in Appendix B, including relaxing the Cobb-Douglas assumption or allowing factor intensities to vary by country. We provide evidence in favor of the Cobb-Douglas assumption for management and overall conclude that the results in a more general setup are if anything stronger than those in Table 4.

5 Conclusion

Developing countries are characterized by a dual economy: large, productive modern firms using new technologies co-exist with small, unproductive traditional firms or own-account workers who produce using out-of-date technologies. We provide evidence that a high cost of management help explain the persistence of the traditional firms. Middle and upper managers are important factor input for large, leading firms and are also expensive in developing countries, with typical salaries over $50,000 per year. These facts deter the expansion of the modern sector through either adoption of new technologies by local firms or expansion of multinational firms.

Our data are not well-suited to test why firms face high costs for management, but we see two leading hypotheses for future research to explore. First, high prices could reflect that leading firms hire particularly skilled workers and such workers are scarce in developing countries – particularly since workers with middle management skills can and do migrate from developing countries on net (Barro & Lee, 2013; Docquier & Rapoport, 2012; Kerr *et al.*, 2016; Martellini *et al.*, 2024). Alternatively, firms could choose to pay higher wages for the same labor, for example in the form of an efficiency wage given weak contract enforcement or in response to intra-firm global pay considerations (Boehm & Oberfield, 2020; Hjort *et al.*, 2020).

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